

Question	Answer	Marks
4(a)	${}^4_2\alpha$	B1
	${}^{211}_{82}\text{Q}$	B1
4(b)(i)	sum / total momentum (of a system of bodies) is constant or sum / total momentum before = sum / total momentum after	M1
	for an isolated system / no (resultant) external force	A1
4(b)(ii)	$p_{\alpha} = p_{\text{P}} - p_{\text{Q}}$ $4(u)v = 215(u) \times 3.2 \times 10^5 (-0)$ $v = 215(u) \times 3.2 \times 10^5 / 4(u)$	C1
	$v = 1.7 \times 10^7 \text{ m s}^{-1}$	A1

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